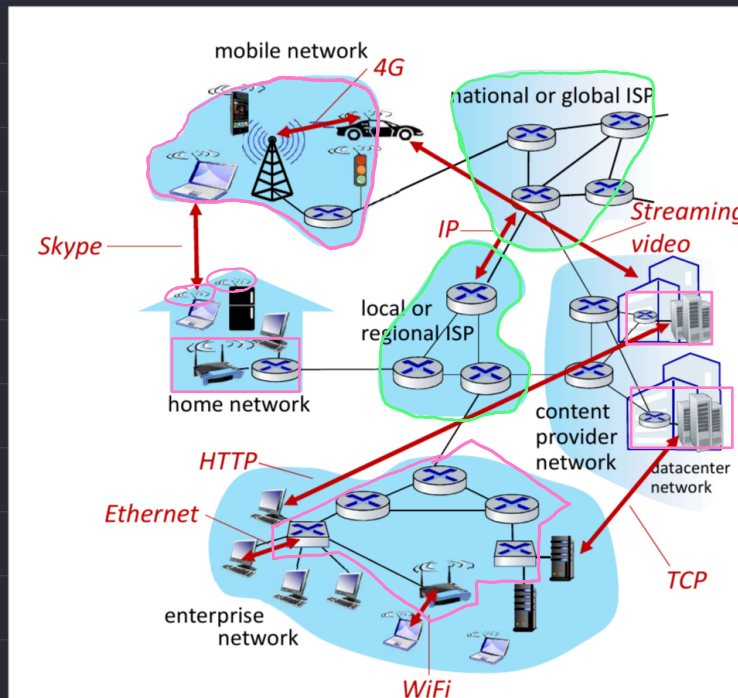


1.3 Network Edge

Infrastructure: 基礎建設

網路使用 API (Programing Interface) 來達成應用程式的資料傳送



End Devices:

Running app on edge



Packet Swticher



Communication links



Networks

Protocol (通訊協議) 定義資料傳送的格式與順序，還有收到資料時的反應方式

Network (Core) 核心交換設備中，不會接到任何的 device，只會連接其他 router / network

Access Network 傳輸端的 Access Network: First mile Network

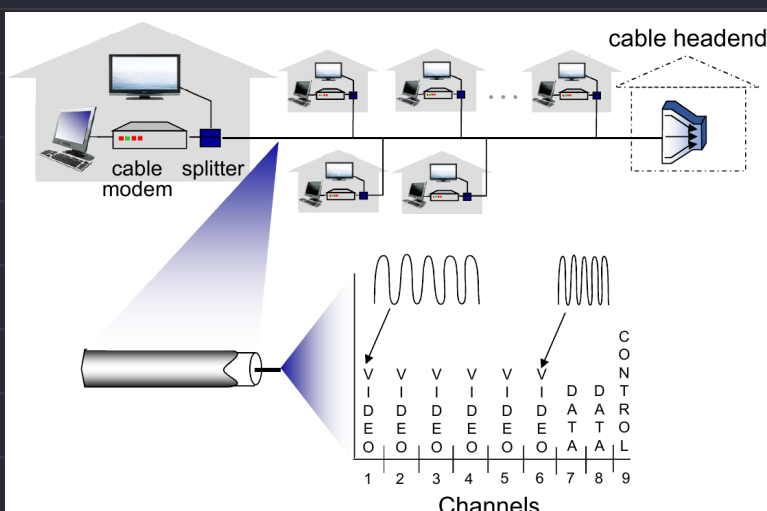
接收端的 Access Network: Last mile Network

Network Edge All devices connected to the internet (including server)

What's a protocol?

設定資料傳輸的格式與順序

Access networks: cable-based access



Cable 網路，所有人共享一個頻寬 (Shared)

FDM (frequency division multiplexing)

用帶通濾波器過濾對應載波頻率的訊號

a.k.a. 不同 channel

HFC (hybrid fiber coax)

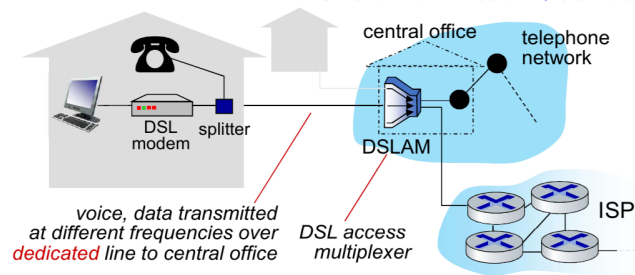
光纖與同軸電纜混合提升傳輸速度

Network of cable

家庭使用線路連到頭端 (headend)

Access networks: digital subscriber line (DSL)

DSL: 有專線直接連到機房, 自己獨享 (Dedicate)



- use **existing** telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

Links (傳輸介質)

Guided Media (有線): Copper, Fiber, Coax

Unguided Media (無線): Radio

Twisted Pair (TP) (雙絞線): CAT5 100Mbps~1Gbps, CAT6 10Gbps

Coaxial Cable (同軸電纜):

Two convertric (同心) copper conductor

Bidirection

Multiple frequency channels on cable (100Mbps per channel)

Fiber optic cable (光纖電纜)

Glass fiber carrying light pulse, each pulse a bit

High speed P2P (10~100Gbps)

Low error rate (immune to electromagnetic noise) + repeaters spaced far apart

1.4 Network Core (Packet / Circuit Switching / Physic Media)

The network core

Hosts break an message into multiple packets, and break packets into bits to trans

$$\text{Packet Transmission Delay} = \frac{L(\text{bits})}{R(\text{bits/sec})}$$

Two key network-core functions

Forwarding: local action, move packet from input to apporiate output link

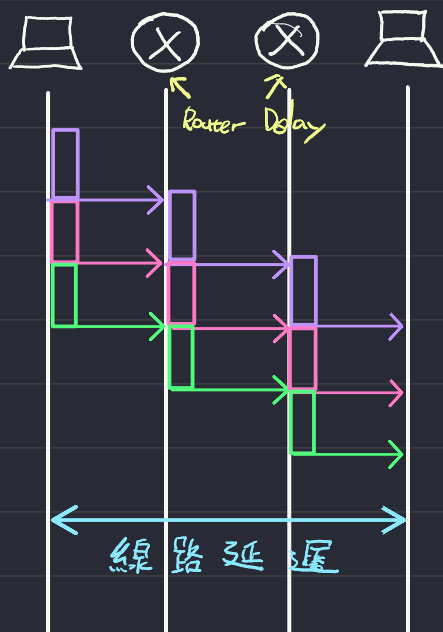
Routing: global action, determine source-destination paths taken by packets
(routing algorithms)

Alternative to packet switching: circuit switching

建立專線，同時能有一個人使用該線路

Packet Switch v.s. Circuit Switch:

Packet Switch:



Circiut Switch:



資料形式	切成封包	原始資料直接傳輸
連線延遲	Router 處理封包需要時間	撥號開始與結束時建立專線（需要時間）
共用	大家隨時都能用	同時只有一能用
人多時	封包遺失、轉發延遲	全部佔線後無法供其他人服務
適用場景	bursty（時有時無） packet	傳統電話

Improvement of Packet Switching

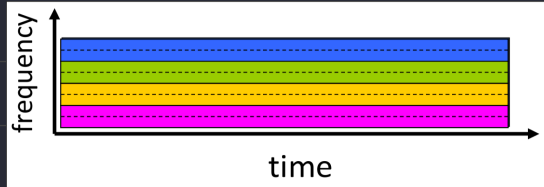
通過 congestion control（壅塞控制）來減少 packet delay 與 buffer overflow

通過加大 bandwidth 來達到 circuit switching 的體驗

Improvement of Circuit switching: FDM / TDM

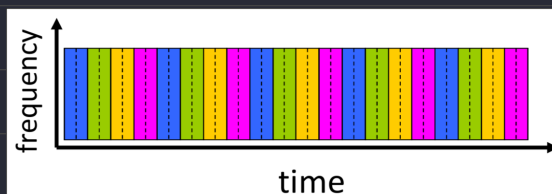
FDM (Frequency Division Multiplexing):

光學或電子信號分成不同頻率，每個訊號可以在該頻率不斷傳輸



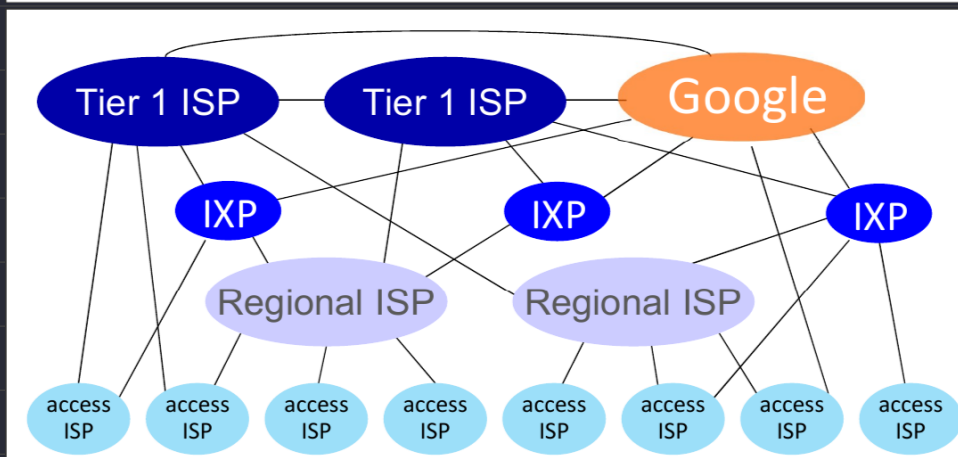
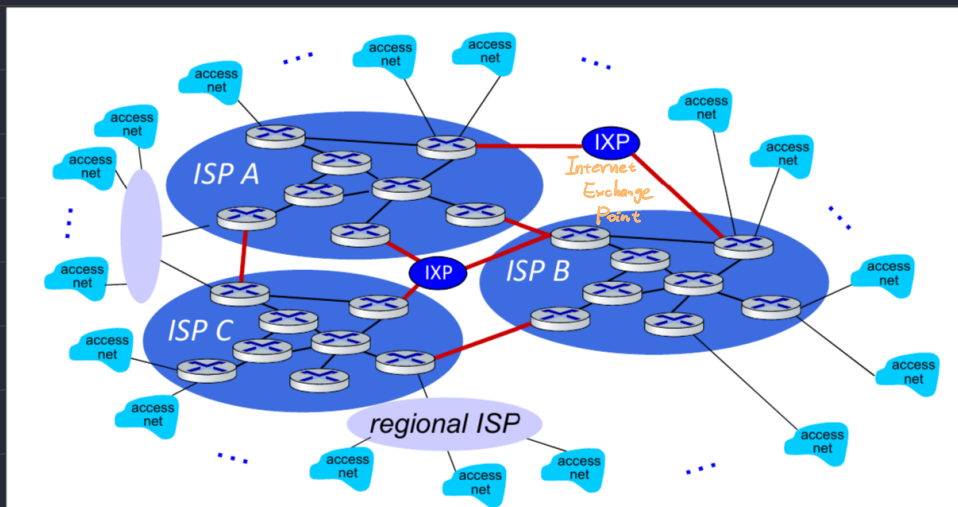
TDM (Time Division Multiplexing):

將時間劃分成時槽(Periodic Slots), 每個訊號輪流發送



Internet structure: a “network of networks”

There are too many access network around the world, and not possible to connect each of them together: $O(n^2)$



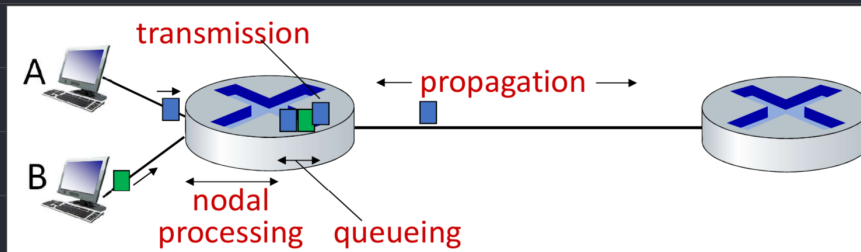
OK 1.5 Performance (Loss / Delay / Throughput)

Packet loss

Arrival rate to link (temporarily) exceeds output link capacity: packet loss

Mostly because the queue of the router has full with packet

Packet delay: four sources



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{proc} : 封包位元錯誤檢查+決定傳輸目的 (可預測的值)

d_{queue} : 在 queue 等待時間, 等 port free (需要用機率預測, 沒有固定值)

d_{trans} : Packet length / Transmission rate (bps)

d_{prop} : Length of physical link / propagation speed ($\sim 2 \times 10^8$ m/s)

在線材中傳輸大概是 2/3 光速

寬頻網路 = 高速網路 $\rightarrow d_{trans}$ 比較大 (因為 rate 大) d_{prop} 不會改變

Packet Queueing delay

R: Link bandwidth (bps)

L: Packet length (bits)

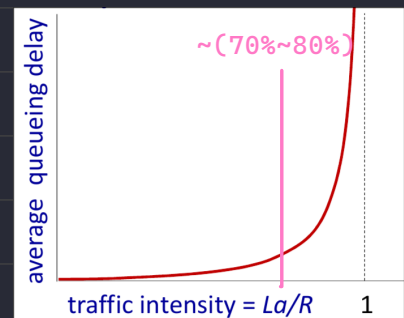
a: Average packet arrival rate (packet/s)

$$L \times a = \text{Bits sent per second}$$

$\frac{La}{R} \approx 0$: Queueing delay small

$\frac{La}{R} \rightarrow 1$: Queueing delay large

$\frac{La}{R} \geq 1$: Queueing delay infinite (Input > Output)



Throughput

Rate (bits/time unit) at which bits are being sent from sender to receiver

- instantaneous (瞬間的): rate at given point in time
- average: rate over longer period of time

Bottleneck link

Throughput of end-end link depends on the smallest one

OK 1.6 Security

Denial of service (DOS)

Attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic

Packet interception (Packet "sniffing")

Promiscuous network interface reads/records all packets (e.g., including passwords!) passing by

Fake identity

IP spoofing: send packet with false source address

(Fake victim's IP e.g. in header to send requests to the server)

OK 1.7 Protocol layers, Service models

Internet protocol stack

application	Support network application (IMAP, SMTP, HTTP) 負責處理特定應用程式的網路協定
transport	Process-process data transfer (TCP, UDP) 兩個應用程式之間建立與確保端對端的連線
network	Routing datagrams from source to destination (IP, routing portocol) 網路層負責在不同的網路之間進行資料的路由和轉發，決定資料從來源主機到目的主機的更佳路徑
link	Data transfet between neighboring network elements (Ethernet, Wifi) 兩個裝置之間傳輸資料
physical	bits (on the wire) 負責透過實體的傳輸媒介傳輸資料

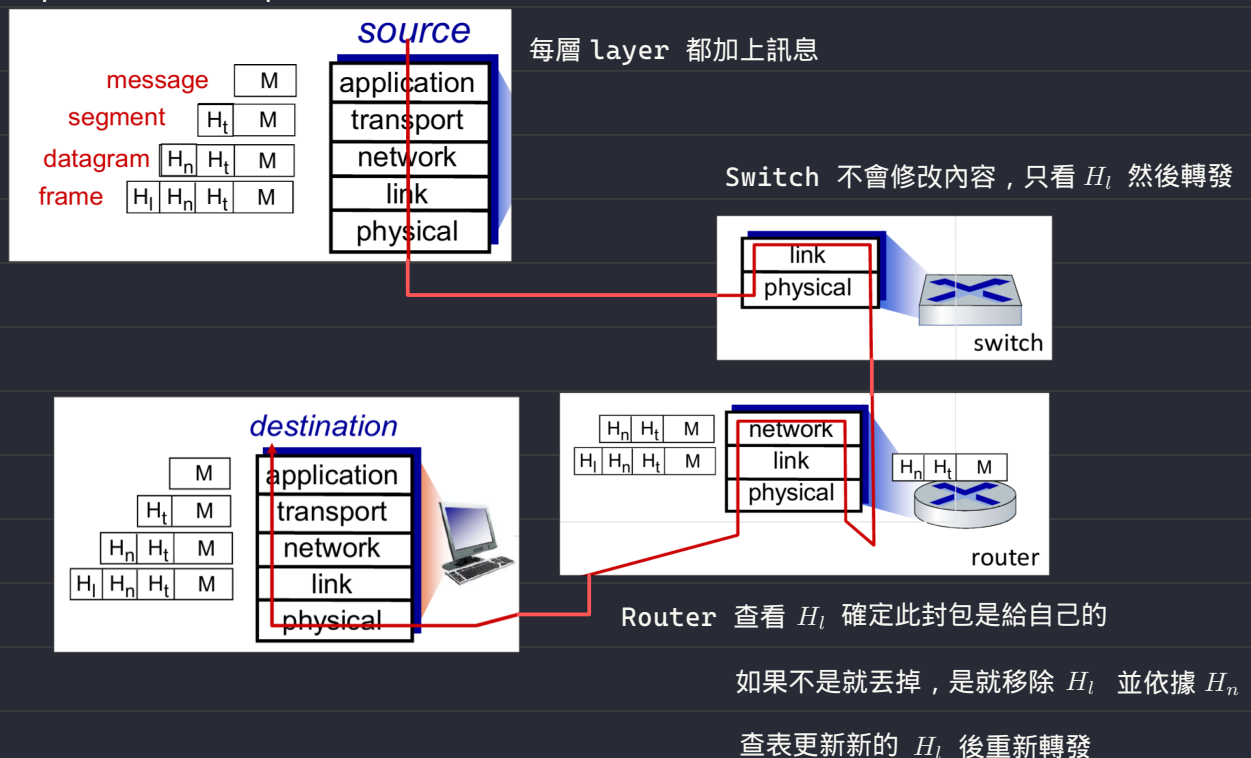
ISO/OSI reference model

現在 Internet 只有五層，但是 OSI 建議要有七層，

而 Presentation、Session 實作在 Internet 的 Applcation Layer

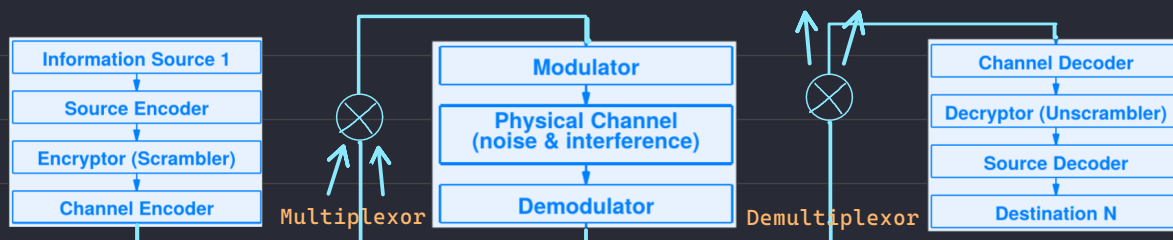
application	presentation: allow applications to interpret meaning of data (e.g., encryption, compression, machine-specific conventions)
presentation	
session	session: synchronization, checkpointing, recovery of data exchange
transport	
network	
link	
physical	

Encapsulation/Decapsulation



OK CH 1.1 Foundations Of Data Communications And The Physical Layer

Motivation And Model



Source Encoder: 壓縮資料簡短傳送的 bit 數量

e.g. Run length encoding $\rightarrow$ AAAABBBCC A4B2C3

Encryptor: 加密資料

Channel Encoder: 減少資料被雜訊干擾而導致的錯誤

e.g. 一個 bit 傳 3 次, 101 $\rightarrow$ 111 000 111, 如果收到 101 $\rightarrow$ 多數決 $\rightarrow$ 1

tradeoff: 犧牲多少速度, 換取多少資料準確度

Modulator: 調變 / 解調變, 分析傳輸環境後, 將訊號調整成最不容易失真的形式傳輸

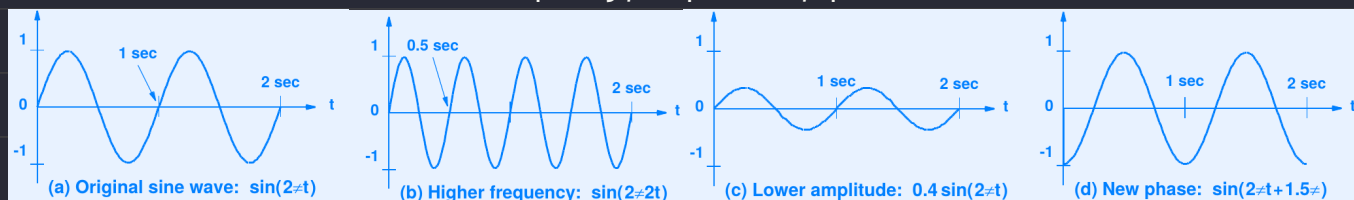
Information Sources And Signals

Fourier Analysis

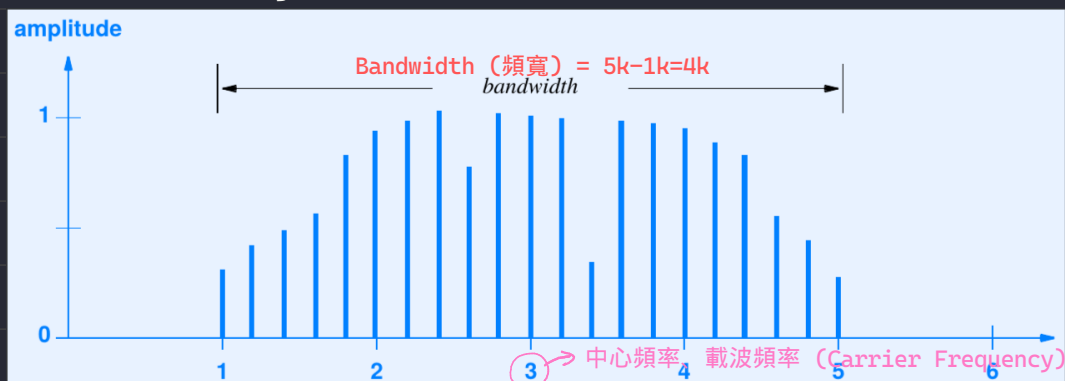
Decompose arbitrary(隨意) wave into multiple sine waves

Digital signals decomposes into an infinite set of sine waves

Sine wave Characteristics: frequency, amplitude, phase



Definition Of Analog Bandwidth

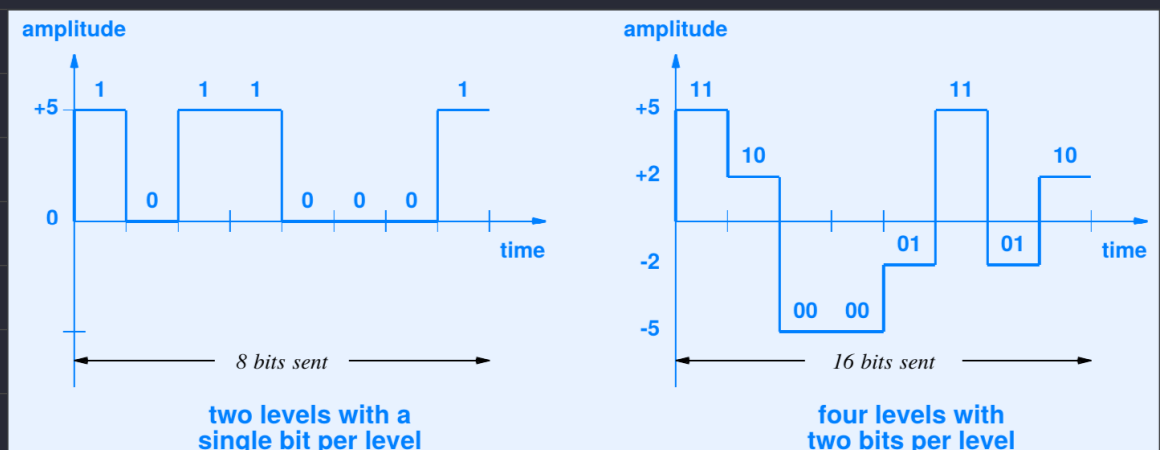


以上訊號是弦波經由傅利葉轉換後得出的圖 $<1k$ || $>5k$ 有可能也有小訊號, 但是可以忽略不計

手機 WiFi 的 2.4Ghz 代表載波頻率, 不代表只用 2.4Ghz 傳輸

Digital Signals And Signal Levels

通過改變振幅種類數量可以決定一次傳輸多少訊號

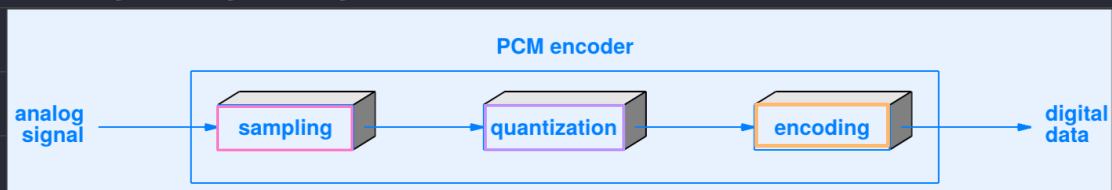


Baud rate (Signal rate) is number of times signal changes per second

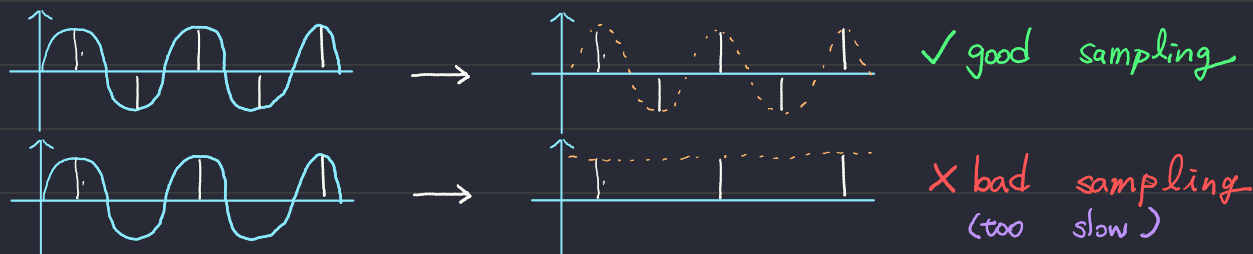
$$\text{data rate in bits per second} = \text{baud} \times \log_2(\text{levels})$$

理論上藉由切成無線多階可以加速傳輸，但實際上要考慮到訊號衰減與雜訊，導致不可行

Converting Analog To Digital



(1) Sampling 取樣



★ 根據 Nyquist Theorem : 取樣頻率至少 = $f_{max} \times 2$

(2) Quantization 量化

將取樣的結果轉換到數位可以接受的大小 e.g. Sample 4.3V → Digital 4.0V

(3) Encoding 編碼

將量化的數值轉成可以被傳輸的訊號

Sampling Rate And Nyquist Theorem

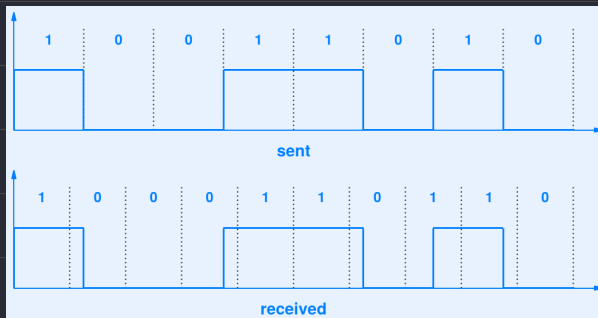
$$\text{sample rate} \geq f_{max} \times 2$$

$$\text{data rate} = 8000 \frac{\text{samples}}{\text{second}} \times 8 \frac{\text{bits}}{\text{sample}} = 64,000 \frac{\text{bits}}{\text{second}}$$

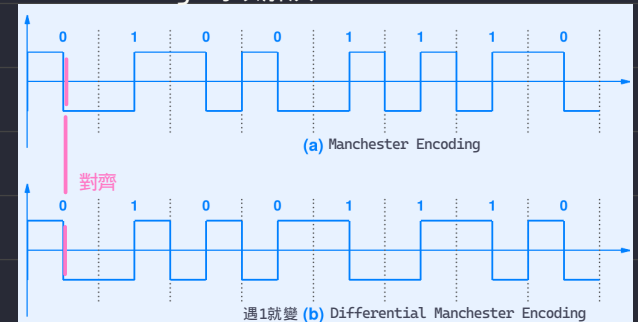
人聲的頻率最高在 4kHz，所以使用 $4k \times 2 = 8kHz$ 的 Sampling rate

Synchronization Errors And Line Coding

發生在發送端與接收端 CLK 不同步的時候



Line Coding 可以解決



Transmission Media

A Taxonomy (分類) Of Transmission Media

能量種類:	Electrical	Light	Electromagnetic
	Twisted Pair (雙絞線)	Optical Fiber	Terrestrial Radio (無線電)
	Coaxial Cable (同軸電纜)	InfraRed (紅外線)	Satellite
		Laser	

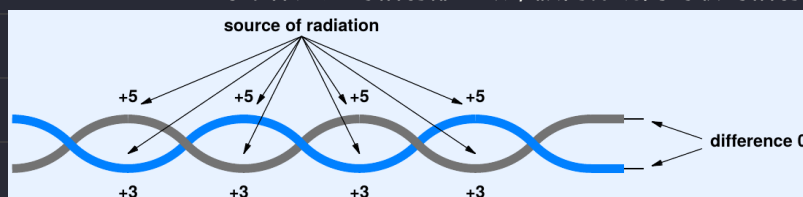
Loss, Interference (電感), And Electrical Noise

Resistance $\leftrightarrow$ loss Capacitance $\leftrightarrow$ distortion Inductance $\leftrightarrow$ interference

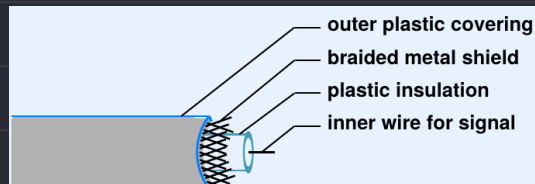
Random electromagnetic radiation $\leftrightarrow$ noise

How Can We Reduce The Effect Of Noise On Copper Wiring

Twisted Pair 可確保經過的輻射都一致，讓兩條現同時收到輻射



Coaxial Cable And Shielding: 在雙絞線外面加其他材料，抗干擾

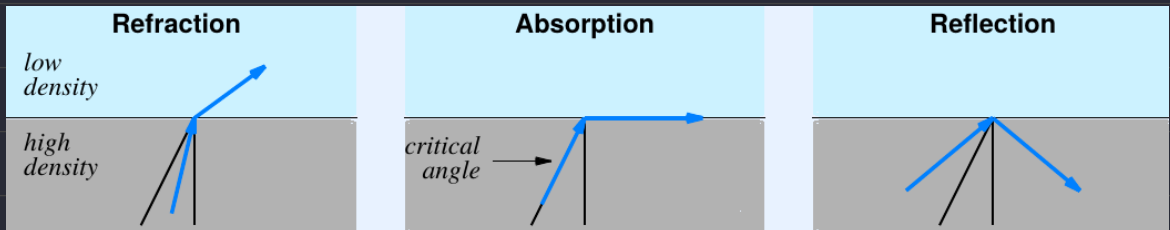


Media Using Light Energy

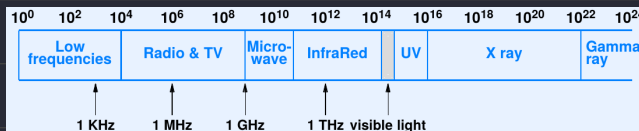
InfraRED transmission (short range and low data rate)

Point-to-point lasers (useful between buildings)

Optical fiber (high data rate and long distance) -- 光纖透過全反射讓光待在裡面



Electromagnetic Spectrum And Properties



Classification	Range	Type Of Propagation
Low Frequency	< 2 MHz	Wave follows earth's curvature, but can be blocked by unlevel terrain
Medium Frequency	2 to 30 MHz	Wave can reflect from layers of the atmosphere, especially the ionosphere
High Frequency	> 30 MHz	Wave travels in a direct line, and will be blocked by obstructions

無線傳輸中有可能因為反射而導致 multipath情況(即電磁波經由多方向反射與延遲而讓接收端收到各種訊號)

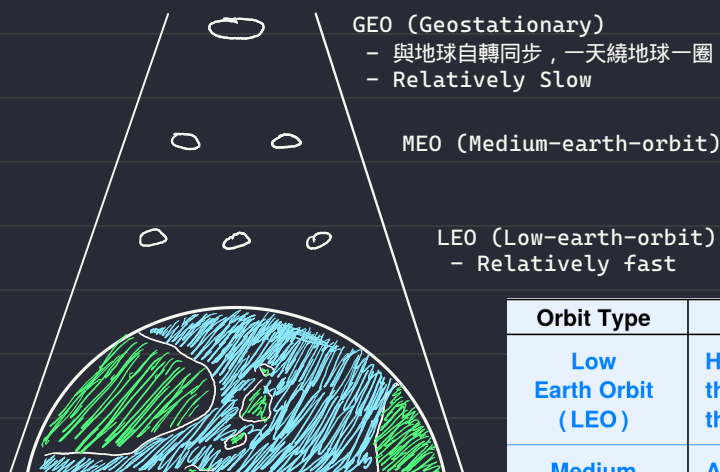


Obstruction by object

Interference (干擾)

Reflection

Satellite Communication



GEO (Geostationary)
- 與地球自轉同步，一天繞地球一圈
- Relatively Slow

MEO (Medium-earth-orbit)

LEO (Low-earth-orbit)
- Relatively fast

Orbit Type	Description
Low Earth Orbit (LEO)	Has the advantage of low delay, but the disadvantage that from an observer's point of view on the earth, the satellite appears to move across the sky
Medium Earth Orbit (MEO)	An elliptical (rather than circular) orbit primarily used to provide communication at the North and South Poles
Geostationary Earth Orbit (GEO)	Has the advantage that the satellite remains at a fixed position with respect to a location on the earth's surface, but the disadvantage of being farther away

Measures Of Transmission Media

Propagation delay

- time required for a signal to traverse a medium

Channel Capacity

- Nyquist's Theorem gives theoretical bound on maximum data rate for hardware bandwidth B and K signal levels
無考慮雜訊

$$\text{Transmission Speed } D = 2 B \log_2 K$$

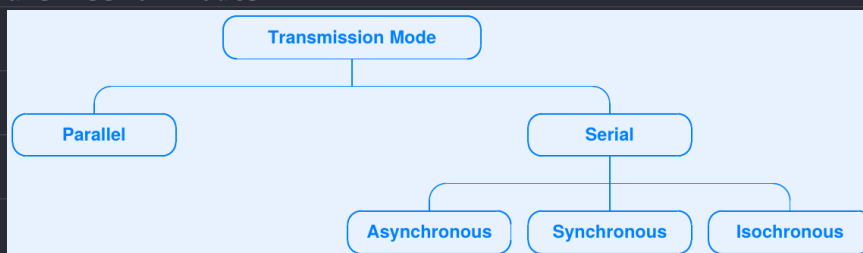
- Mathematical result known as *Shannon's Theorem* gives the maximum channel capacity, C , in the presence of noise
有考慮雜訊

$$C = B \log_2(1 + S/N)$$

雜訊比

- Quantity S/N is known as the *signal-to-noise ratio*

Transmission Modes



Ethernet uses byte big-endian and bit little-endian order

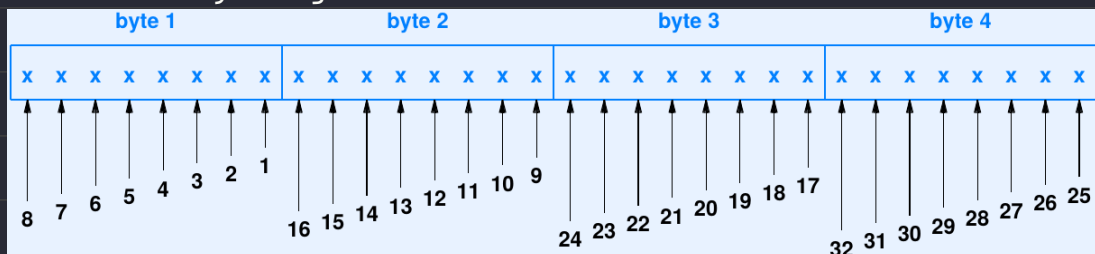
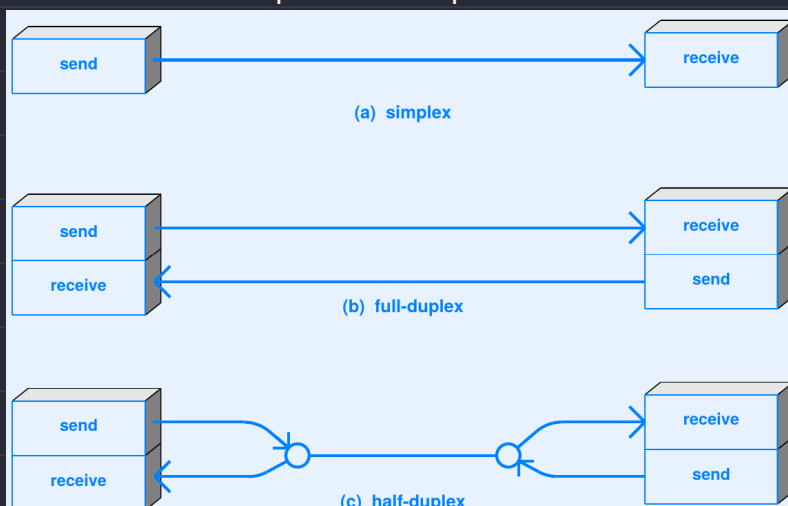
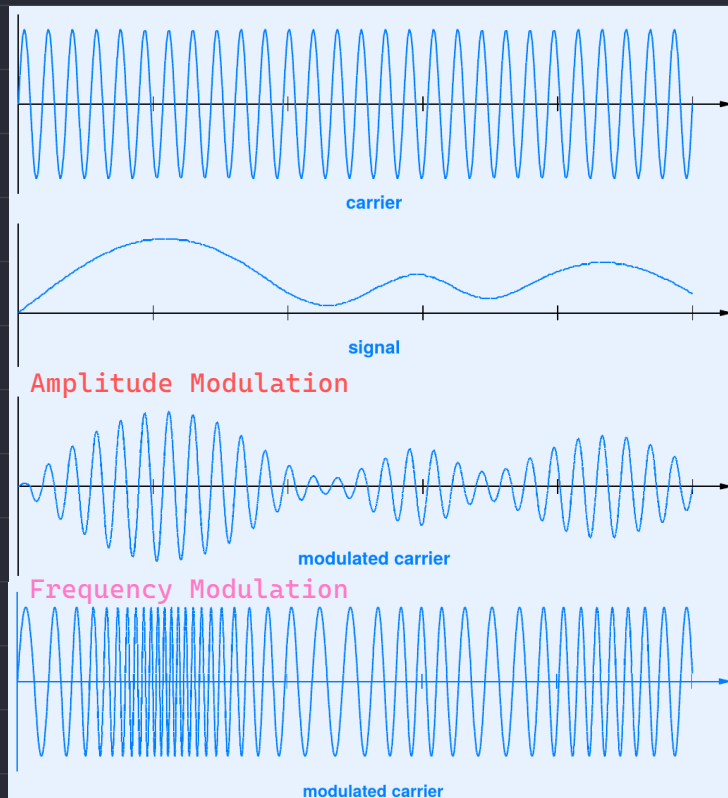


Illustration Of Simplex And Duplex Modes



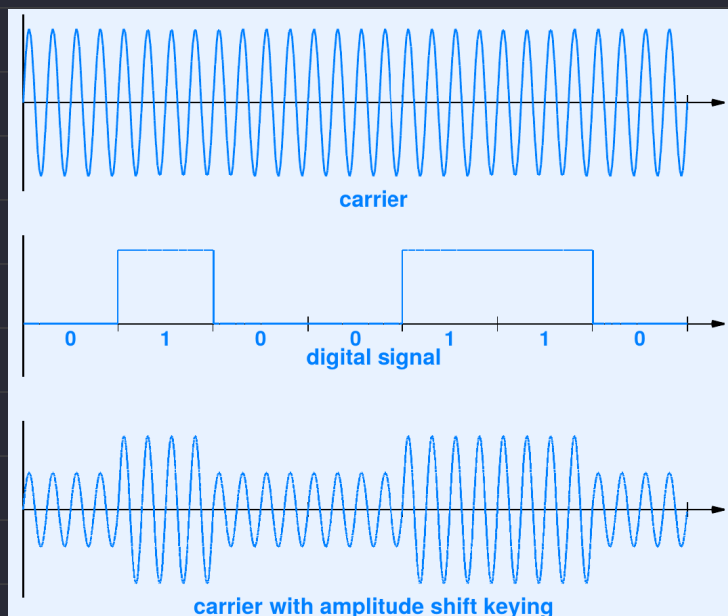
Modulation And Demodulation



Shift Keying

→ Like modulation but input signal is digital

ASK (amplitude Shift Keying)

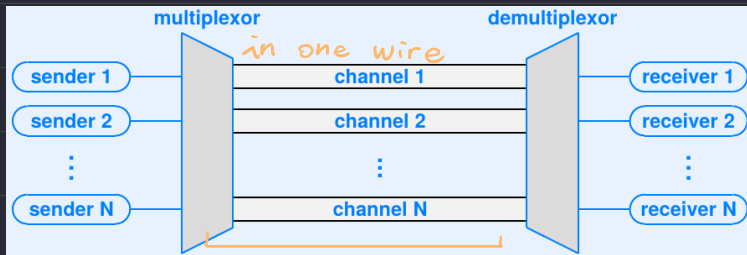


PSK (Phase shift keying)

改變 sine wave 的相位角，然後用代通濾波器解調變

將 ASK + PSK = QAM (一次可代表 4 種訊號 e.g. Wifi)

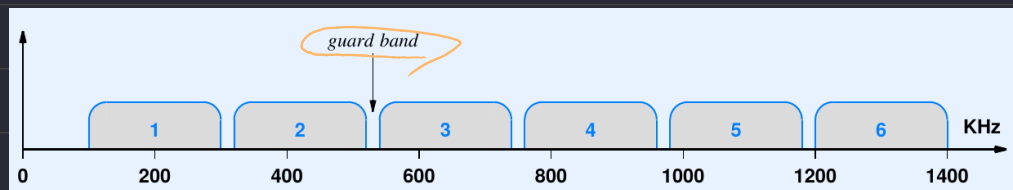
Multiplexing And Demultiplexing (Channelization)



分頻方式

1. FDM (Frequent Division Modulation)

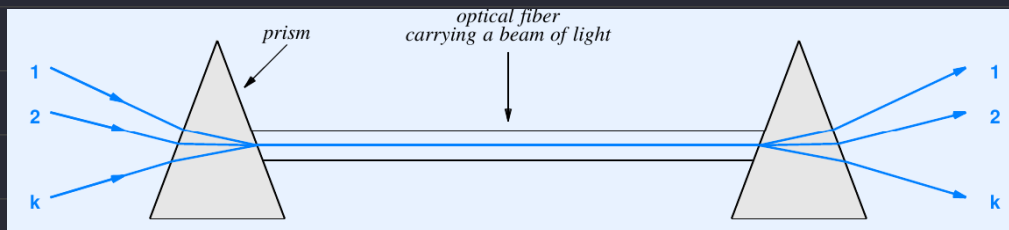
藉由載波頻率 (Carrier Frequency) 不同來使用代通濾波器分開訊號



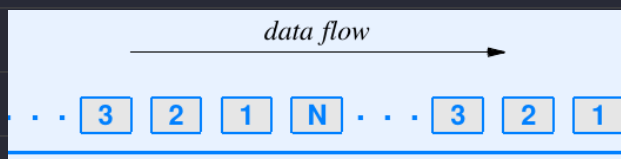
2. Wavelength Division Multiplexing (WDM) :

通過三稜鏡分散/集合光的不同頻率來傳輸資料

在光纖中使用 Dense WDM (DWDM) 每個 channel 有 10Gbps 的帶寬



3. Time Division Multiplexing



Synchronous TDM: 時間一到就給特定 channel 傳輸, 無論是否有需求

Statistical TDM: 如果設備有需求才配傳輸 slot (時槽)

4. Code Division Multiplexing

- 1) Each sender/receiver pair is assigned a unique number called a chip sequence
- 2) Senders multiply the data value by their chip sequence (orthogonal vector s
- 3) Transmitted value is a sum of all senders
- 4) Each receiver multiplies incoming value by its chip sequence to extract dat

在低負載時有低延遲, e.g. Cellphone

5. Hierarchical Multiplexing (階層式多工)

Used with FDM and TDM to combine multiple **lower-capacity** channels

- Example of TDM hierarchy used by the phone system

